



# HOW TO CONSTRUCT AN NEURAL PATH



► Can think of the annealing path as a continuous corruption of the target  $\pi$  to noise  $\eta$

$$\eta(\mathrm{d}x) = \eta(x)\mathrm{d}x, \quad \pi(\mathrm{d}x) = \pi(x)\mathrm{d}x = \frac{\gamma(x)}{Z}\mathrm{d}x$$

►  $\beta \mapsto \gamma_\beta(x)$  is a continuous function for all  $x \in X$



$\gamma_0 = \eta$  and  $\gamma_1 = \gamma$

- We build annealing distribution by corrupting the densities:

$$\pi_{\beta}(x) = \frac{\gamma_{\beta}(x)}{Z(\beta)}, \quad Z(\beta) = \int_{\mathbb{X}} \gamma_{\beta}(x) dx$$



▶ We build an neural distribution by capturing the densities:

$$V_{\beta} \equiv \log \gamma_{\beta}, \quad A(\beta) \equiv \log Z(\beta)$$



$Z(\beta) < \infty$  for all  $\beta \in [0, 1]$

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  - ▶  $\gamma_0 = \eta$  and  $\gamma_1 = \gamma$
  - ▶  $Z(\beta) < \infty$  for all  $\beta \in [0,1]$
- ▶ We build annealing distribution by corrupting the densities:

$$V_\beta = \log \gamma_\beta, \quad A(\beta) = \log Z(\beta)$$

# LINEAR PATH